

**I ranked products  
by total\_sold\_quantity  
LIMIT 3 kept one division  
two divisions vanished**

**ATLIQ HARDWARE GDB0041**

**MYSQL | Vishnu Ram Sachin | Data Analyst**



# Problem Statement

DAY 10 / 15

05/08/26

- Yesterday : Created 5 CTE for readability and added other deductions
- Leadership wants top 3 products by quantity sold in each division **P & A, PC , N & S**

## The Trap ❌

- Sorted every product by total\_sold\_quantity then LIMIT 3

Only 3 products return

All products from PC , P& A vanished



# Requirment

DAY 10 / 15

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**PARTITION BY division**



**ORDER BY total DESC**



**Split the rows first ,  
then rank inside the split**

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## The Fix

**Use Window Function**

**RANK () OVER(PARTITION BY ORDER BY DESC)**



# A Quick Analogy

DAY 10 / 15

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## Exam Result

### One merit list

every student ranked against the whole school  
( ORDER BY + LIMIT )

### Split by section

10 - A , 10 - B, 10 - C graded  
( PARTITION BY division )

### Podium per event

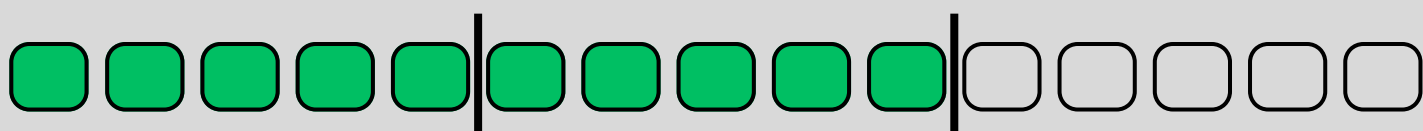
each section produces its own toppers  
( RANK ( ) OVER ( ) )

### Final report card

3 sections x 3 toppers = 9 names  
( Final SELECT )



```
WITH ranked_products_report AS (  
  SELECT  
    dp.product_code , dp.product , dp.division ,  
    SUM(fsm.sold_quantity) total_sold_quantity,  
    RANK() OVER (PARTITION BY division ORDER BY SUM(fsm.sold_quantity) DESC )  
  AS product_rank  
  FROM fact_sales_monthly fsm  
  JOIN dim_product dp  
    ON fsm.product_code = dp.product_code  
  GROUP BY dp.division , dp.product_code )  
  
SELECT  
  rpp.product_code , rpp.product ,  
  rpp.division , rpp.total_sold_quantity, rpp.product_rank  
  FROM ranked_products_report rpp  
  WHERE product_rank <=3  
  ORDER BY rpp.division , rpp.product_rank
```



# Results

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**fact\_sales\_monthly** | **dim\_product** | SQL File 10\*

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Limit to 50000 rows

```

1 • WITH ranked_products_report AS (
2     SELECT
3         dp.product_code , dp.product , dp.division ,
4         SUM(fsm.sold_quantity) total_sold_quantity,
5         RANK() OVER (PARTITION BY division ORDER BY SUM(fsm.sold_quantity) DESC ) AS product_rank
6     FROM fact_sales_monthly fsm
7     JOIN dim_product dp
8         ON fsm.product_code = dp.product_code
9     GROUP BY dp.division , dp.product_code )
10
11    SELECT
12        rpp.product_code , rpp.product ,
13        rpp.division , rpp.total_sold_quantity, rpp.product_rank
14    FROM ranked_products_report rpp
15    WHERE product_rank <=3
16    ORDER BY rpp.division , rpp.product_rank

```

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Result Grid | Filter Rows: | Export: | Wrap Cell Content:

|   | product_code | product                  | division | total_sold_quantity | product_rank |
|---|--------------|--------------------------|----------|---------------------|--------------|
| ▶ | A6720160103  | AQ Pen Drive 2 IN 1      | N & S    | 1332238             | 1            |
|   | A6818160201  | AQ Pen Drive DRC         | N & S    | 1300534             | 2            |
|   | A6319160201  | AQ Neuer SSD             | N & S    | 1207499             | 3            |
|   | A2118150106  | AQ Master wired x1 Ms    | P & A    | 1216615             | 1            |
|   | A2118150105  | AQ Master wired x1 Ms    | P & A    | 1210759             | 2            |
|   | A2219150203  | AQ Master wireless x1 Ms | P & A    | 1210271             | 3            |
|   | A4218110202  | AQ Digit                 | PC       | 56791               | 1            |
|   | A4118110107  | AQ Aspiroon              | PC       | 56363               | 2            |
|   | A4218110201  | AQ Digit                 | PC       | 56149               | 3            |

Result 4 ×



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**I Vishnu Ram Sachin I Data Analyst**

**SQL THROUGH ONE DATASET CHALLENGE**